PROJECT UPDATE: AI WORKLOAD PROFILING ON A RISC-V CORE

Fatimata Abdou Dramane

MOTIVATION AND GOAL

Why profile an AI workload?

- The Hardware-Software Gap
- Kernel Bottlenecks
- The Need for Visibility

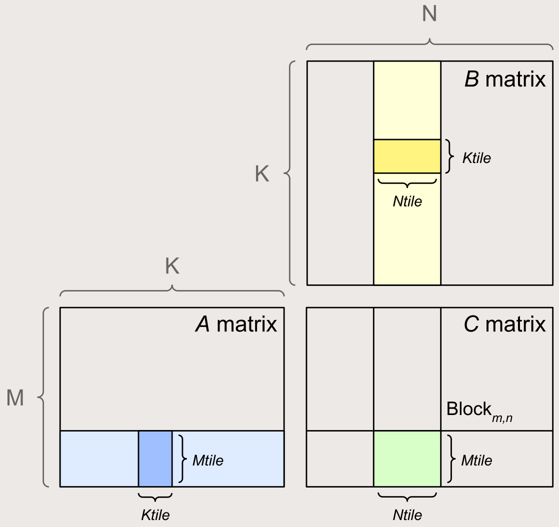
How this project help?

- Build an End-to-End Profiling Pipeline
- Identify Bottlenecks
- Drive Hardware-Software Co-Design

BENCHMARKING WITH GEMM & SIMULATING WITH GEM5

Why GEMM?

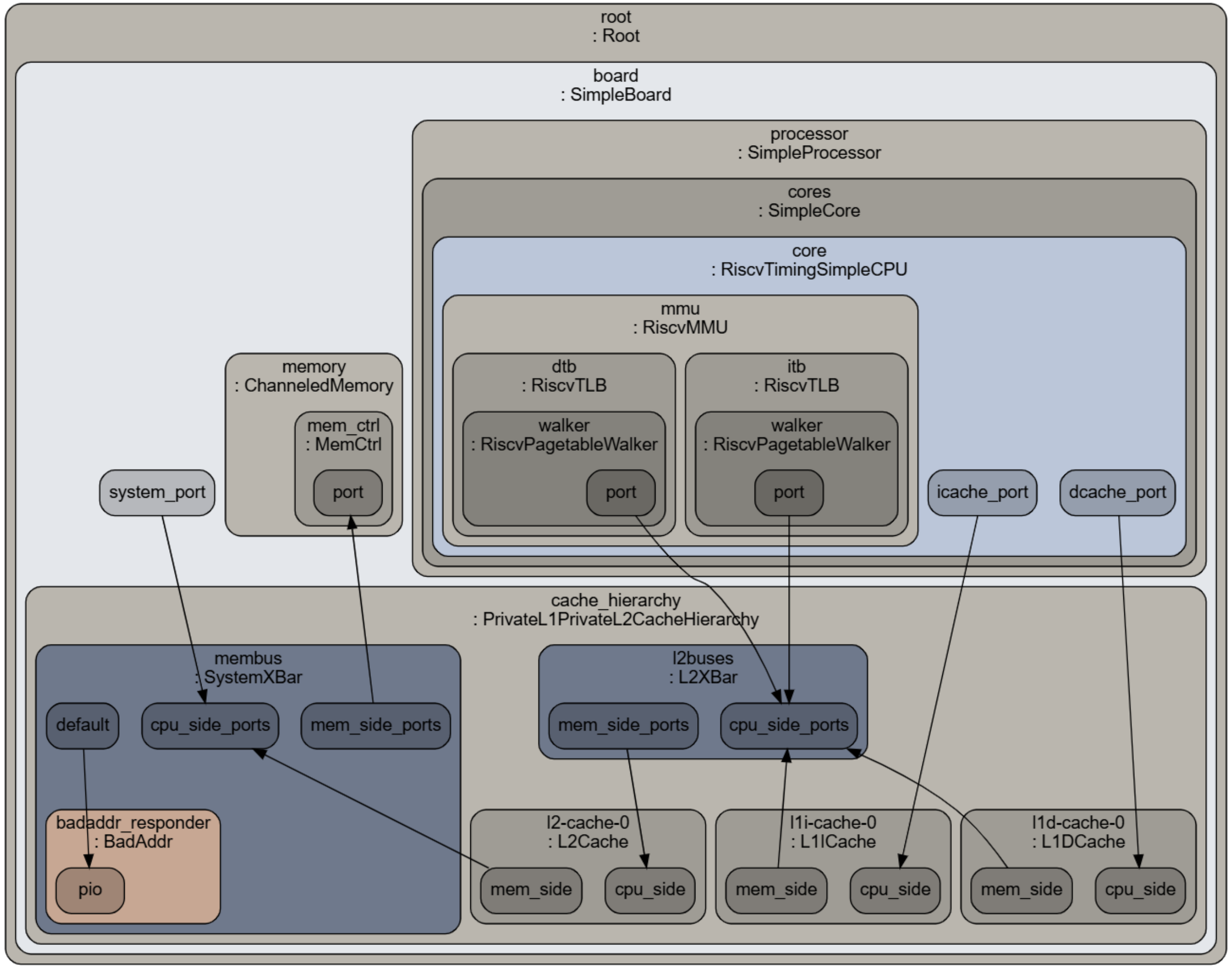
- The core mathematical operation (C = A x B) powering modern AI models, machine learning algorithms, and LLMs.
- Dominates execution time and heavily taxes processor resources.
- Easy to verify, highly optimized, and acts as the ultimate benchmark for processor limits.



Source: nvidia docs

WHY GEM5?

- Simulates physical hardware behavior step-by-step.
- Measures statistics you can't easily see on real boards, like cache misses, execution cycles, and memory stalls.
- Eliminates the need for expensive, hard-to-find physical RISC-V development boards.

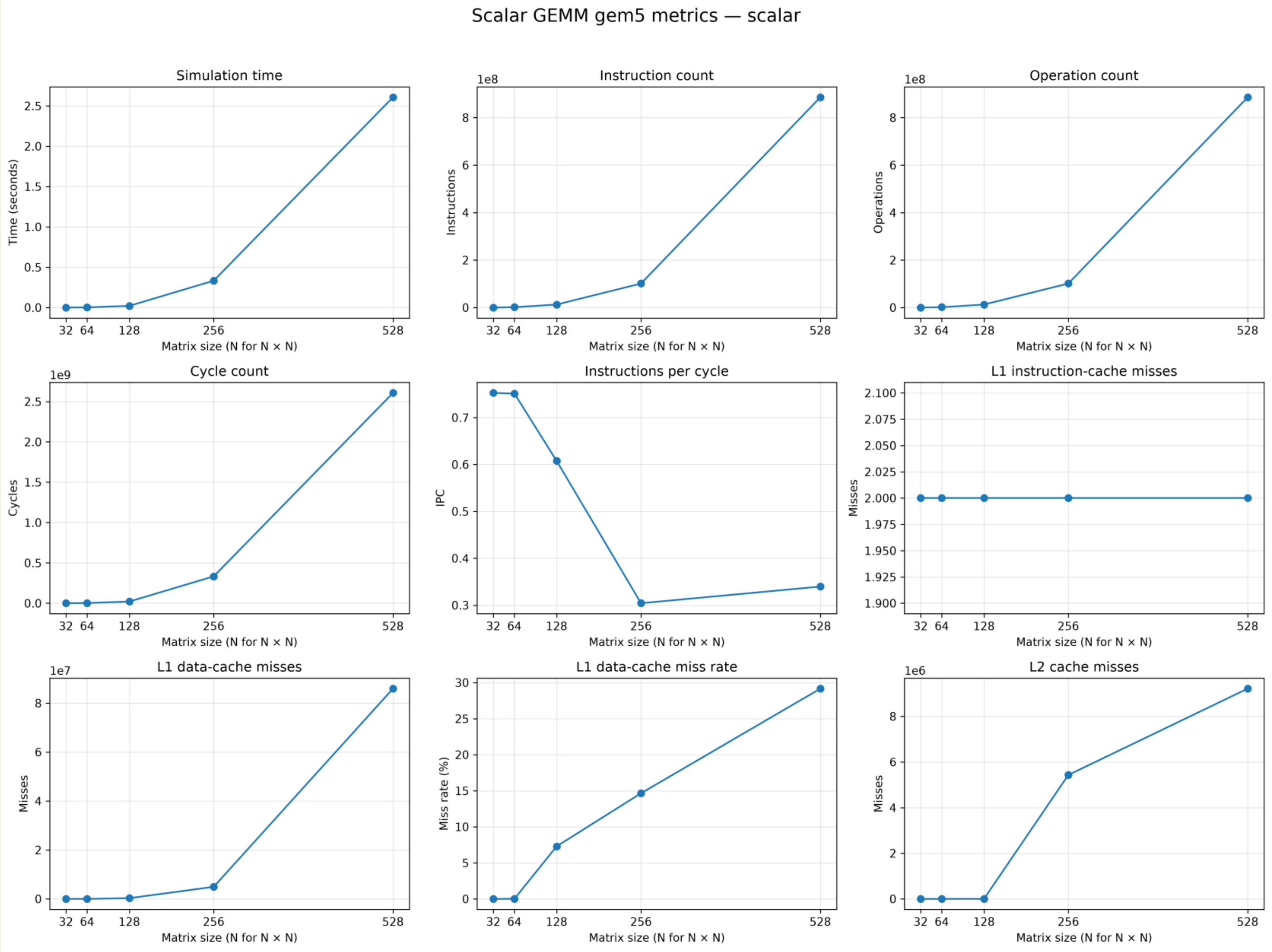


Source: Gem5 documentation

MILESTONES ACHIEVED

- **Algorithms:** Coded scalar GEMM and verified mathematical correctness.
- **Analysis:** Evaluated assembly differences between unoptimized (`-O0`) and optimized (`-O2`) compilation.
- **Simulator:** Integrated code with gem5, added Region of Interest (ROI) markers, and automated statistic parsing and plotting.

AI WORKLOAD PROFILING ON A RISC-V CORE



THE PATH FORWARD

Next Steps (Operators & Vectors)

- Port GEMM to RISC-V Vector Extensions (RVV).
- Implement and profile other critical AI operators (e.g., ReLU, LayerNorm).

The Ultimate Goal (Model-Level Reasoning)

- **Full-Scale WorkloadsPredictive Performance:** Link high-level user metrics (e.g., model size, sequence length, input tokens) to physical hardware behavior.

THANK YOU

workload-profiling

riscv-ottawa / **Profile-on-RISC-V**